

CareSync

AI-Powered Remote Healthcare Platform

3-Month Development Plan

Development Team	
Backend & DevOps:	Ramsey
Frontend & UI/UX:	Abbie
Timeline:	3 Months (12 Weeks)
Target:	Pilot Launch with Healthcare Agencies

Project Overview

CareSync is an AI-powered remote healthcare platform focused on elderly and chronically ill patients. The platform enables continuous care from home through health data logging, remote monitoring, video consultations, and AI-generated insights.

Core Features	
Health Data Logging	BP, glucose, weight, temperature, O2 levels, symptoms
Remote Monitoring	Real-time dashboards for doctors and caregivers
Video Consultations	Built-in video calls via Daily.co integration
AI Summaries	Automated consultation summaries and trend analysis

Smart Alerts	Rule-based notifications for critical readings
Appointment System	Scheduling and automated email reminders

Technology Stack

Component	Technology	Why
Backend	Python FastAPI	Fast, modern, type-safe
Database	PostgreSQL	Structured health data, HIPAA-ready
Frontend	Next.js + React	Modern, fast, SEO-friendly
Styling	Tailwind CSS	Rapid UI development
Video	Daily.co API	10k free minutes/month
AI	OpenAI GPT-4o-mini	Cost-effective summaries
Hosting	Railway/Render + Vercel	Free tiers available
Email	Resend/SendGrid	Free tier for alerts

MONTH 1: Foundation & Core Features

Weeks 1-4: Authentication, Patient Management, Health Data Logging, Dashboard & Alerts

Week 1: Setup & Authentication

Backend Tasks (You):

- Project setup (FastAPI + PostgreSQL)
- Database schema design
- User authentication system (JWT)
- Multi-tenant setup (agency isolation)
- User roles: Admin, Doctor, Caregiver, Patient
- Basic CRUD endpoints for users
- Docker setup for local development
- Environment configuration

Frontend Tasks (Abigael):

- Next.js project setup
- Tailwind CSS configuration
- Design system/component library (shadcn/ui)
- Login/Register pages
- Basic routing structure
- API client setup (axios/fetch)
- Authentication state management

Deliverable: Working auth system with role-based access

Week 2: Patient Management

Backend Tasks (You):

- Patient model & database schema
- CRUD endpoints: Create, List, Get, Update, Delete patients
- Patient-caregiver-doctor relationships
- Pagination & filtering
- Input validation (Pydantic models)

Frontend Tasks (Abigael):

- Patient list page (dashboard)
- Add new patient form

- Patient detail view
- Edit patient form
- Search & filter UI
- Responsive mobile design (caregivers use phones!)

Deliverable: Complete patient management (CRUD)

Week 3: Health Data Logging

Backend Tasks (You):

- Health readings model (BP, glucose, weight, temp, O2, heart rate)
- Flexible schema for different reading types
- Endpoints: Log readings, Get history, Get latest vitals
- Photo/file upload (medication bottles, wounds)
- Validation rules (BP range, glucose range)
- Trend calculation logic

Frontend Tasks (Abigael):

- Health data entry form (quick & detailed modes)
- Photo capture/upload
- Reading history view
- Trend graphs using Recharts (BP, glucose, weight over time)
- Mobile-optimized input

Deliverable: Health logging with graphs

Week 4: Dashboard & Alerts

Backend Tasks (You):

- Alert system architecture
- Rule-based alerts (high/low BP, glucose, missed readings)
- Alert endpoints (List, Acknowledge)
- Email notification system (SendGrid/Resend)
- Alert prioritization (critical, warning, info)

Frontend Tasks (Abigael):

- Doctor/Caregiver dashboard with patient list
- Alert feed (critical alerts highlighted)
- Quick stats (total patients, alerts today, consultations)
- Alert notification UI (toast notifications, alert center)
- Patient status cards (healthy, needs attention, critical)
- Dashboard filters (by risk level, last contact)

Deliverable: Complete dashboard with real-time alerts

MONTH 2: Video & AI Features

Weeks 5-8: Video Consultations, AI Summaries, Scheduling, Admin Panel

Week 5: Video Consultations

Backend Tasks (You):

- Daily.co API integration
- Consultation scheduling model
- Endpoints: Schedule, List, Start (generate room), Complete
- Store consultation metadata (duration, participants, notes)
- Recording webhook handler (if using Daily.co recording)

Frontend Tasks (Abigael):

- Schedule consultation form (date/time picker, select patient/doctor)
- Consultation list view (upcoming & past)
- Video call UI (embed Daily.co, in-call vitals display)
- End call button
- Mobile video optimization

Deliverable: Complete video consultation flow

Week 6: AI Summaries (YOUR DIFFERENTIATOR!)

Backend Tasks (You):

- OpenAI API integration (GPT-4o-mini)
- Consultation summary generation (notes + vitals → summary)
- Weekly health summary (week of data → trend analysis)
- Endpoints: Generate consultation summary, Weekly report
- Prompt engineering for medical context
- Error handling & retry logic
- Summary storage in database

Frontend Tasks (Abigael):

- Consultation summary view (display AI summary after call)
- Edit/annotate summary, Save final version
- Weekly report view (auto-generated, trend highlights)
- AI loading states
- Summary history

Deliverable: AI summary generation working

Week 7: Scheduling & Reminders

Backend Tasks (You):

- Appointment scheduling system
- Doctor availability management
- Endpoints: Create, List, Reschedule, Cancel appointments
- Automated reminder system (cron job: 24hrs & 1hr before)
- Email reminders (no SMS yet - too expensive)
- Daily health check-in reminders

Frontend Tasks (Abigael):

- Appointment booking flow (calendar view, time slot selection)
- Appointment management (reschedule, cancel)
- Reminder preferences settings
- Calendar integration (export to Google Calendar)

Deliverable: Scheduling with automated reminders

Week 8: Admin Panel & Agency Management

Backend Tasks (You):

- Agency (tenant) management
- Endpoints: Create agency, Get stats, List patients/users
- Billing/usage tracking (patients, consultations, AI summaries)
- Analytics endpoints (patient growth, consultation frequency, alerts)

Frontend Tasks (Abigael):

- Admin dashboard (agency overview, user management)
- Analytics views (charts for growth, consultations, alerts)
- Usage metrics (for billing discussions)
- Settings panel

Deliverable: Admin panel for agencies

MONTH 3: Polish, Testing & Pilot

Weeks 9-12: Optimization, Testing, Deployment, Launch

Week 9: Mobile Optimization & UX Polish

Backend Tasks (You):

- API performance optimization (indexing, query optimization, caching)
- Rate limiting
- API documentation (FastAPI auto-docs + examples)
- Error handling improvements
- Logging & monitoring setup

Frontend Tasks (Abigael):

- Mobile responsiveness audit (test on real devices)
- Progressive Web App (PWA) setup (offline, add to home screen)
- UX improvements (loading states, error messages, tooltips)
- Accessibility audit (basic)

Deliverable: Mobile-optimized, polished UI

Week 10: Testing & Bug Fixes

Backend Tasks (You):

- Unit tests (auth, alerts, AI summaries)
- Integration tests (API endpoints, database)
- Security audit (SQL injection, XSS, CORS, encryption)
- Load testing (basic)
- Bug fixes from frontend testing

Frontend Tasks (Abigael):

- Manual testing across all flows
- Cross-browser testing
- Mobile device testing
- Create bug list & fix critical bugs
- User acceptance testing (UAT with friends/family)

Together: End-to-end testing, test scenarios, integration bug fixes

Deliverable: Tested, bug-free system

Week 11: Deployment & Documentation

Backend/DevOps Tasks (You):

- Production database setup (PostgreSQL on Railway/Render/Supabase)
- Environment variables management
- Backend deployment (Railway/Render)
- SSL/HTTPS setup
- Database backups setup
- Monitoring (Sentry free tier)
- CI/CD pipeline (GitHub Actions - basic)
- API documentation finalization

Frontend Tasks (Abigael):

- Frontend deployment (Vercel)
- Environment configuration (API URLs)
- Custom domain setup (if available)
- Performance optimization (images, code splitting, lazy loading)
- Analytics setup (PostHog/Plausible free tier)
- User documentation/help section

Together: Full system smoke test, demo account with sample data, pitch deck/demo video

Deliverable: Deployed system on custom domain

Week 12: Pilot Launch & Iteration

Your Tasks:

- Onboard first pilot agency/hospital
- Setup their account & users
- Train administrators
- Monitor system performance
- Quick bug fixes
- Collect feedback
- Plan iteration 2

Abigael's Tasks:

- Create onboarding materials (videos, guides, FAQ)
- User support (respond to questions)
- Track user behavior (most-used features)
- UI tweaks based on feedback
- Iterate on confusing flows

Together: Weekly pilot review meetings, document learnings, prioritize iteration 2

Deliverable: Live pilot with real users!

Success Metrics

Month	Goals
Month 1	• Complete auth & patient management • Health data logging works end-to-end
Month 2	• Video calls working • AI summaries generating • Alert system functional
Month 3	• Live pilot with 1 hospital/agency • 20-50 patients in system • Feedback collected for v2

Tools & Workflow

- **Project Management:** GitHub Projects or Trello (free) - Task tracking
- **Version Control:** GitHub - Code repository
- **Communication:** Discord/Slack - Daily communication
- **Design:** Figma (free) - UI mockups
- **Documentation:** Notion/Google Docs
- **API Testing:** Postman/Insomnia
- **Development:** VS Code, Docker for local environment

Daily Routine

Morning: 15-min async standup (text update on Discord/Slack)

- What you did yesterday
- What you're doing today
- Any blockers

Evening: Push code, update task board

Weekly: 1-hour sync call (discuss progress, plan next week)

Budget Breakdown (Monthly)

Service	Free Tier	Cost When Scaling
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Daily.co (video)	10k min/month	\$24/month (100 patients)
OpenAI (AI)	Pay-as-you-go	~\$3-6/month (MVP)
Railway/Render (backend)	Free tier	\$5-10/month
Vercel (frontend)	Free tier	Free (hobby)
Supabase (database)	500MB free	\$25/month (growth)
SendGrid (email)	100/day free	Free for MVP
Total MVP Cost	\$0-10/month	\$60-75/month (growth)

Risk Mitigation Strategies

If Behind Schedule...	Mitigation Strategy
Weeks 1-4 behind	Skip admin panel (Week 8), focus on core patient features
Weeks 5-7 behind	Simplify AI summaries (use templates instead of full AI)
Weeks 9-12 behind	Launch with known bugs, fix iteratively with pilot feedback
Frontend struggles	Pair programming on complex components, use UI templates (shadcn/ui)
Money runs out	Daily.co free tier → WebRTC Reduce AI summary frequency Use all free hosting tiers

Pro Tips for Success

- **Start with the hardest part first** - Video + AI in Month 2 prevents last-minute crunch
- **Deploy early** - Don't wait till Week 11, deploy by Week 4 to catch issues early
- **Get real users by Week 8** - Even 5 patients from family/friends for testing
- **Document as you go** - Future you will thank present you
- **Don't over-engineer** - Ship features quickly, iterate based on feedback
- **Communicate daily** - 15-min async updates prevent blockers
- **Celebrate small wins** - Deployed auth system? Celebrate! First video call? Celebrate!
- **Ask for help** - Stuck on something for >2 hours? Reach out to communities (Discord, Reddit)

Next Steps

1. Review this plan together and discuss timeline feasibility
2. Set up communication channels (Discord/Slack)
3. Create GitHub repository and project board
4. Design database schema (Week 1, Day 1)
5. Start coding! ■

